Transcript

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In 1901, David Hänig published a paper that forever changed our understanding of taste. His research led to what we know today as the taste map: an illustration that divides the tongue into four separate areas. According to this map, receptors at the tip of our tongues capture sweetness, bitterness is detected at the tongue's base, and along the sides, receptors capture salty and sour sensations. Since its invention, the taste map has been published in textbooks and newspapers. The only problem with this map, is that it's wrong. In fact, it's not even an accurate representation of what Hänig originally discovered. The tongue map is a common misconception— something widely believed but largely incorrect. So where do misconceptions like this come from, and what makes a fake fact so easy to believe?

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It's true that the tongue map's journey begins with David Hänig. As part of his dissertation at Leipzig University, Hänig analyzed taste sensitivities across the tongue for the four basic flavors. Using sucrose for sweet, quinine sulfate for bitter, hydrochloric acid for sour, and salt for salty, Hänig applied these stimuli to compare differences in taste thresholds across a subject's tongue. He hoped to better understand the physiological mechanisms that affected these four flavors, and his data suggested that sensitivity for each taste did in fact vary across the tongue. The maximum sensation for sweet was located at the tongue's tip; bitter flavors were strongest at the back; salt was strongest in this area, and sour at the middle of the tongue's sides. But Hänig was careful to note that every sensation could also be tasted across the tongue, and that the areas he identified offered very small variations in intensity.

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Like so many misconceptions, the tongue map represents a distortion of its original source, however the nature of that distortion can vary. Some misconceptions are comprised of disinformation— false information intentionally designed to mislead people. But many misconceptions, including the tongue map, center on misinformation— false or misleading information that results from unintentional inaccuracy.

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Misinformation is most often shaped by mistakes and human error, but the specific mistakes that lead to a misconception can be surprisingly varied. In the case of the tongue map, Hänig's dissertation was written in German, meaning the paper could only be understood by readers fluent in German and well versed in Hanig's small corner of academia. This kicked off a game of telephone that re-shaped Häing's research every time it was shared with outside parties. Less than a decade after his dissertation, newspapers were falsely insisting that experiments could prove sweetness was imperceptible on the back of the tongue.

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The second culprit behind the tongue map's spread were the images that Hänig's work inspired. In 1912, a rough version of the map appeared in a newspaper article that cautiously described some of the mysteries behind taste and smell research. Featuring clear labels across the tongue, the article's illustration simplified Hänig's more-complicated original diagrams. Variations of this approachable image became repeatedly cited, often without credit or nuanced consideration for Hänig's work. Eventually this image spread to textbooks and classrooms as a purported truth of how we experience taste.

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But perhaps the factor that most contributed to this misconception was its narrative simplicity. In many ways, the map complements our desire for clear stories about the world around us— a quality not always present in the sometimes-messy fields of science. For example, even the number of tastes we have is more complicated than Hänig's work suggests. Umami— also known as savory— is now considered the fifth basic taste, and many still debate the existence of tastes like fatty, alkaline, metallic, and water-like.

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Once we hear a good story, it can be difficult to change how we see that information, even in the face of new evidence. So, next time you see a convenient chart or read a surprising anecdote, try to maintain a healthy skepticism— because misconceptions can leave a bitter taste on every part of your tongue.